



VA-7431

**Integrated M. C. A. (Sem. III)
Examination, December - 2023**

IMCA-301

**OBJECT ORIENTED PROGRAMMING
USING C++**

Time allowed : Three Hours

Maximum Marks : 70

Minimum Pass Marks : 28

*This question paper contains three sections
as under:*

SECTION - A

- 1 (i) Define OOP.
(ii) Name any four OOP languages.
(iii) Explain cout in C++.
(iv) Explain Ternary operator in C++.
(v) Define this pointer in C++.
(vi) Define friend function in C++.
(vii) Explain inheritance.
(viii) Define virtual class.
(ix) Define function template in C++.
(x) Define throw in C++.

SECTION - B

UNIT - I

- 2 Explain the difference between procedural programming and object oriented programming.

OR

Explain various features of OOP.

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[Contd...

UNIT - II

- 3 Explain various data types available in C++.

OR

Write a C++ program to swap two numbers using call by reference.

UNIT - III

- 4 Explain parameterized constructor in C++.

OR

Explain operator overloading with suitable C++ program.

UNIT - IV

- 5 Explain inheritance and its types.

OR

Write a C++ program to implement virtual function.

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[Contd...

UNIT - V

- 6 Write a program using try.....catch block to handle divide by zero exception.

OR

Explain class template with suitable example.

SECTION - C

- 7 (a) Explain Destructor in C++ with suitable example.
(b) Write a program in C++ to add two 2D array using call by value.
- 8 Write a program to calculate square and square root of a floating point number using pointers.

- 9 Explain the concept of polymorphism in C++ using suitable program.
- 10 Explain various loops available in C++.
- 11 Explain class scope under inheritance with suitable example.
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